ResumeScope (AI-Resume-Analyzer)

Complete Project Guide — What It Is, How It Works, & Architecture Explained Simply

Developer: Rishav Raj	**License:** MIT Open Source
GitHub Repository: github.com/rishavrjpt/AI-Resume-Analyzer	**Core Stack:** Python 3.11, FastAPI, Scikit-Learn, PyTorch
Web App: http://127.0.0.1:8000 (Swagger: /docs)	**Automated Tests:** 28/28 Pytest Test Cases Passing (100%)

1. What is this Project?

ResumeScope is a smart, transparent personal resume review application. When applying for jobs, candidates struggle to know whether their resume clearly highlights the key skills requested by employers, whether bullet points are specific, and which important technical terms are missing from their document.

ResumeScope takes your resume (in PDF, Word .docx, or plain text format) and compares it with a real job description. It uses **Natural Language Processing (NLP)** and **Machine Learning (ML)** to extract sections, identify verified skills, calculate honest lexical and semantic alignment percentages, and provide evidence-grounded suggestions to make bullet points stronger.

> **Core Philosophy:** This is an empowering tool for the job seeker — **NOT** an applicant screening system. It never predicts whether someone should be hired, never produces fake 'ATS pass scores', and never invents qualifications.

2. Why is it Different from Typical Resume Checkers?

Commercial resume review sites frequently use deceptive practices or basic keyword counts. ResumeScope fixes this:

- **Zero Fake 'ATS Scores':** Commercial tools show made-up numbers like '79% ATS Score'. In reality, ATS systems do not use a universal scoring algorithm. ResumeScope reports honest wording similarity and semantic alignment instead.
- **Zero Hallucinated Content:** Generic LLM chatbots often invent fake experience or fake metrics. ResumeScope uses safe template placeholders like *'[add measured result if you have one]'*.
- **100% In-Memory Privacy:** Uploaded files are parsed in RAM and immediately released. No documents are stored on disk, and phone numbers/emails are automatically masked.
- **Honest Term Labeling:** Unmatched keywords are labeled politely as *'Not found in the provided resume text'*, recognizing that the user might have the skill without explicitly mentioning it.
- **Fast Local Processing:** All NLP components run locally without sending your private career history to third-party commercial cloud APIs.

3. Step-by-Step NLP Pipeline

When a user submits a resume and target job description, the system executes an automated 5-stage pipeline in under 2 seconds:

Step	Pipeline Component	What it Does in Simple English
1	Document Parsing	Accepts PDF, DOCX, or TXT. Extracts clean text in memory, strips corrupt characters, and masks Indian phone numbers and emails for privacy.
2	Section Classification	Scans text lines and identifies standard sections (Summary, Skills, Experience, Education, Projects, Certifications) using trained machine learning.
3	Skill Extraction	Matches technical terms against a transparent taxonomy of 250+ skills and catches capitalized domain acronyms (e.g., UPI, REST, JWT, RBAC).
4	Dual Matching	Compares resume with job description using two methods: Lexical TF-IDF (word overlap) and SentenceTransformers MiniLM (deep semantic meaning).
5	Grounded Feedback	Audits resume bullet points for weak verbs (e.g., 'helped with') and unquantified metrics, providing non-inventing template rewrites.

4. Machine Learning Models & Real Benchmark Results

ResumeScope was evaluated on strictly held-out test data with zero data leakage across document boundaries. Here are the genuine evaluation results:

Model Task	Architecture / Technique	Metric	Baseline	ResumeScope	Improvement
Section Detection	TF-IDF (1-3 n-grams) + Logistic Regression	Macro F1 Score	0.4181	0.8215	+0.4034 (+96.5%)
Skill Extraction	Boundary-aware Taxonomy + Acronym Filter	F1 Score	0.6800	0.9825	+0.3025 (+44.5%)
Lexical Matching	TF-IDF Cosine Similarity	Spearman Correlation	0.5200	0.8932	+0.3732 (High Correlation)
Semantic Matching	SentenceTransformers all-MiniLM-L6-v2	Spearman Correlation	0.4800	0.8870	+0.4070 (High Correlation)

5. Tailored for the Indian Tech Ecosystem

ResumeScope was deliberately designed with domain knowledge of the Indian software and engineering hiring market:

- **4 Major Indian Tech Hub Roles:** One-click presets for **Bengaluru** (Senior Backend & Fintech), **Gurugram** (Product & Data Analytics), **Hyderabad** (Applied AI & NLP), and **Pune** (Modern Frontend React/Next.js).
- **Indian Academic Degrees:** Natively recognizes B.Tech, B.E., M.Tech, BCA, MCA, and Indian university grading (e.g., 'CGPA: 8.7/10.0').
- **Core High-Demand Tech Terms:** Built-in taxonomy recognizes DSA (Data Structures & Algorithms), System Design, Microservices, Spring Boot, UPI / Payment Webhooks, Redis In-Memory Caching, and Kafka Streaming.

6. The ML Testing Dashboard & Synthetic Resumes

A hallmark of a serious AI project is empirical testing. ResumeScope features a dedicated **ML Testing Dashboard** with **8 synthetic test resumes** representing distinct real-world situations:

Sample Filename	Persona & Tech Hub	Edge Case Tested
resume_01_software_engineer.txt	Rahul Sharma (Bengaluru Backend)	Standard headings, dedicated skills section, high-concurrency microservices.
resume_02_data_analyst.txt	Pooja Verma (Gurugram Analyst)	Data analytics, SQL, Tableau, Power BI, Python data wrangling.
resume_03_ml_intern.txt	Sneha Kulkarni (IISc ML Intern)	Academic & research projects, Hugging Face Transformers, PyTorch.
resume_04_frontend_developer.txt	Ananya Iyer (Chennai Frontend)	Modern web dev: React, Next.js, Redux, Tailwind CSS, Jest unit testing.
resume_05_backend_developer.txt	Aarav Patel (Pune Backend)	Skills mentioned inside experience bullets with NO separate skills heading.
resume_06_cybersecurity_trainee.txt	Priya Shukla (Hyderabad Security)	Penetration testing, Wireshark, CEH certification, network security.
resume_07_project_manager.txt	Vikramaditya Rao (Noida PM)	Agile methodologies, sprint management, stakeholder alignment, non-coding.
resume_08_minimal_resume.txt	Aditi Sen (Kolkata Junior)	Sparse resume with missing sections to verify graceful fallback handling.

7. How to Install and Run Locally

Anyone can run this project on their computer in three simple steps:

Step 1: Clone repo and install requirements:

```
git clone https://github.com/rishavrjpt/AI-Resume-Analyzer.git
cd AI-Resume-Analyzer
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
```

Step 2: Run automated test suite:

```
pytest tests/ -v (Runs all 28 automated tests in ~3.8 seconds)
```

Step 3: Launch web application:

```
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload
```

Open **http://127.0.0.1:8000** in your browser. Click any of the 4 Indian sample resumes, or upload your own resume, and click **Analyze Document**!

8. Frequently Asked Questions (Interview & Viva Guide)

Q: Why did you use TF-IDF + Logistic Regression instead of calling an LLM?

A: It gives sub-5ms latency, 100% deterministic output, zero cost per inference, requires no network access, and achieves an 0.82 Macro F1 score on document-isolated test sets.

Q: How does the system handle special technical symbols like C++, C#, and Next.js?

A: Standard regex word-boundary anchors (\b) fail on punctuation like '+', '#', and '.'. ResumeScope uses custom boundary lookarounds that preserve symbols while preventing false substring matches.

Q: What is the difference between Lexical and Semantic similarity in this project?

A: Lexical (TF-IDF Cosine) checks exact technical vocabulary overlap. Semantic (all-MiniLM-L6-v2) projects texts into a 384-dimensional vector space to recognize that 'PostgreSQL' and 'relational database' share meaning even without sharing characters.

Q: How did you prevent data leakage during model training?

A: Dataset splitting was enforced strictly at the document level (document_id). All sentences from any single resume belong entirely to either train, validation, or test — never split across sets.

Q: What makes this project compliant with Responsible AI principles?

A: It never outputs hireability scores, never predicts hiring success, scrubs PII upon ingestion, does not retain resumes on disk, and formats suggestions using non-inventing template placeholders.